Moral reasoning project

Small presentation

Plan

1. Motivation and inspirations
2. Early draft of a framework
3. Some experiments and ideas

1. Motivation and Inspirations

Why moral reasoning for M-Rat?

- **Same initials (MR).**
- A compelling case of Multi-Perspectivality.
- Also:
  - Growing interest related to rapid improvement of LLMs.
  - Some nice resources from Philosophy.

Taking moral reasoning seriously

1. No clear ground truth (vs logic or science)
    - Pluralism & Asystematicity of this normative domain.
    - E.g., conflict between different values in an ethical dilemma.
    - But still some reasonings are better than others!

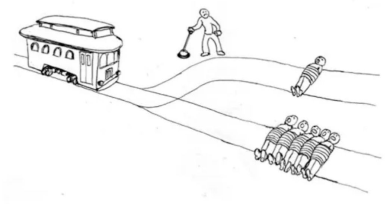
2. May admit multiple uncomparable different solutions.
    - Depending on values or moral framework.

Sources of inspiration

1. Reflective equilibrium
2. Multi-objective optimization

Reflective equilibrium (RE)

A.



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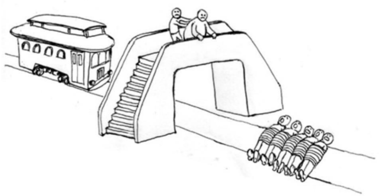
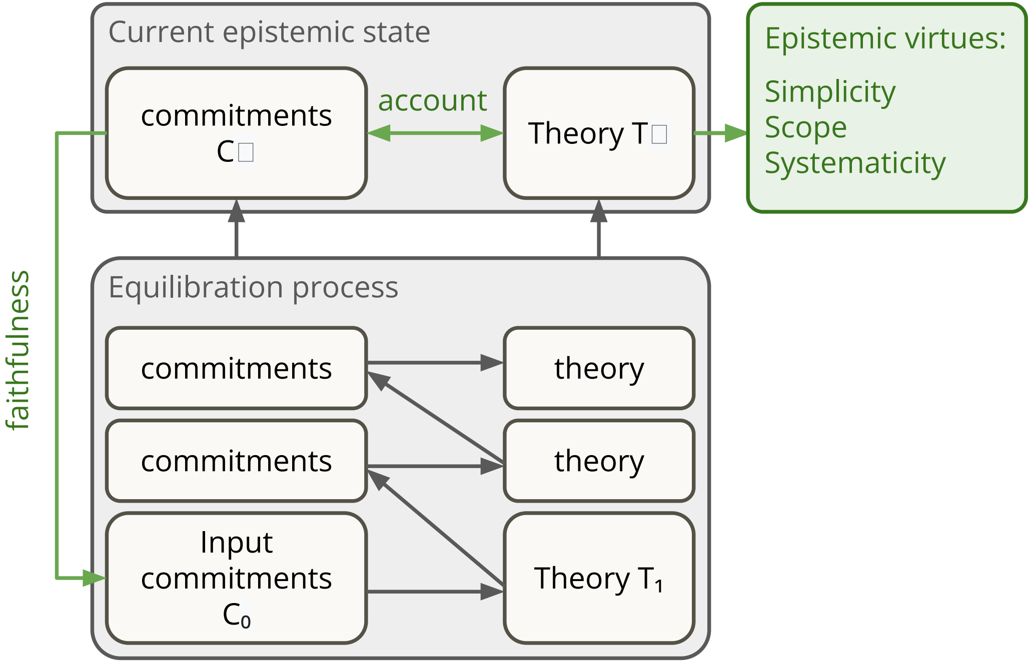


Commitments:
Pull in A.
Push in B.

Commitments:
Pull in A
Don’t push in B.

Principle:
Killing 1 to save 5 is ok.

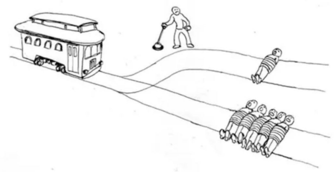
A formal model of RE



Claus Beisbart, Making Reflective Equlibrium Precise: A Formal Model, 2020, Ergo

A blueprint for LLM-MR?

A.



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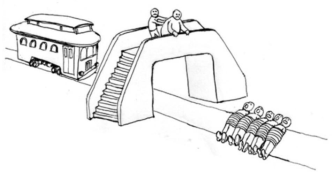
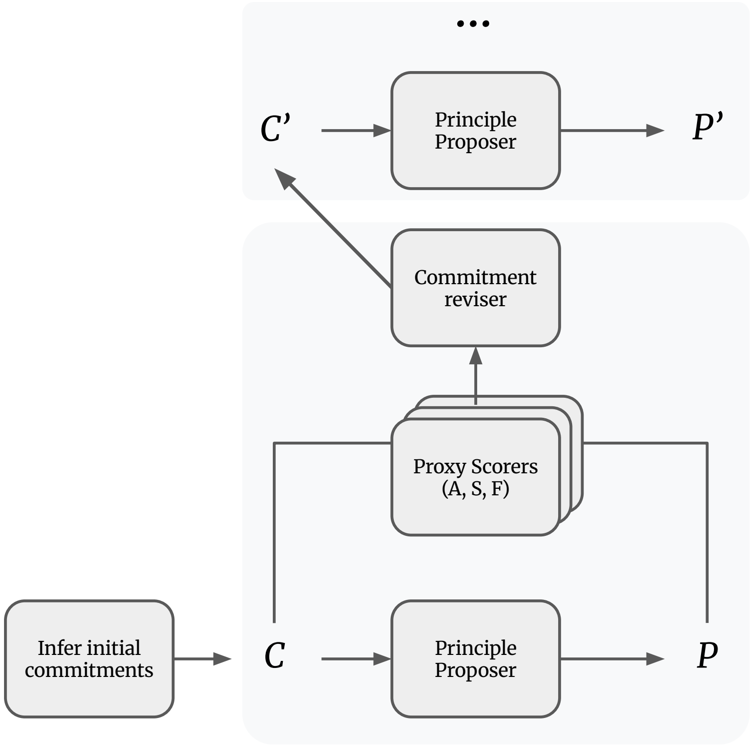
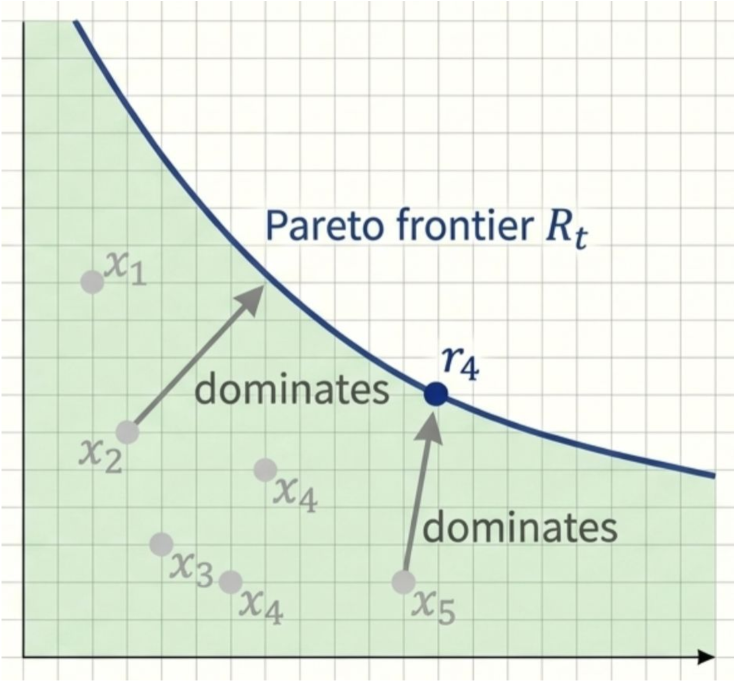




Multi-objective optimization



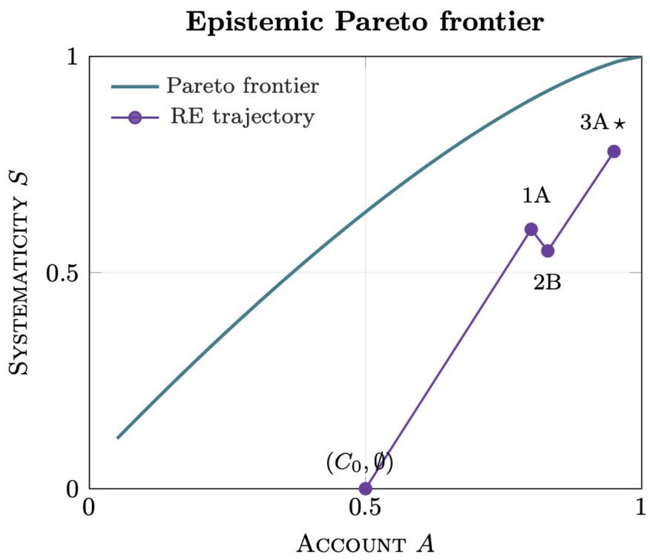
A **Pareto front** represents the set of solutions where no solution outperforms any other solution in the set at every objective.

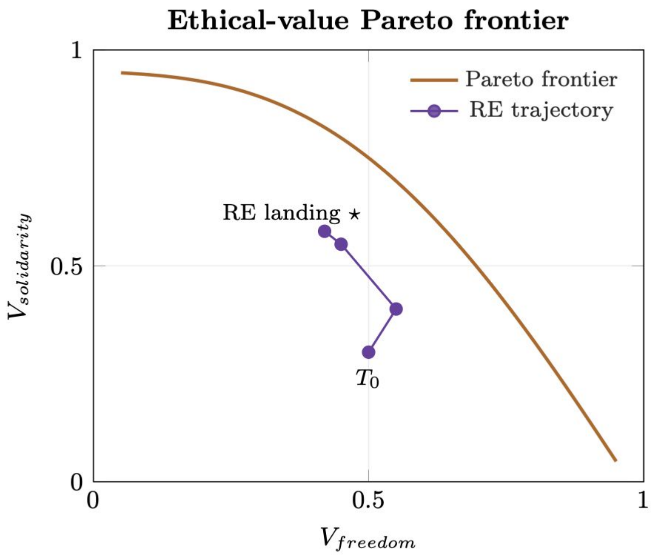
→ Potentially a good tool to model the pluralism of the moral domain!

Task is to find a dominating solution (on the front). Or maybe even multiple!

Using this allows us to **move away from "gold standard" benchmarking**.

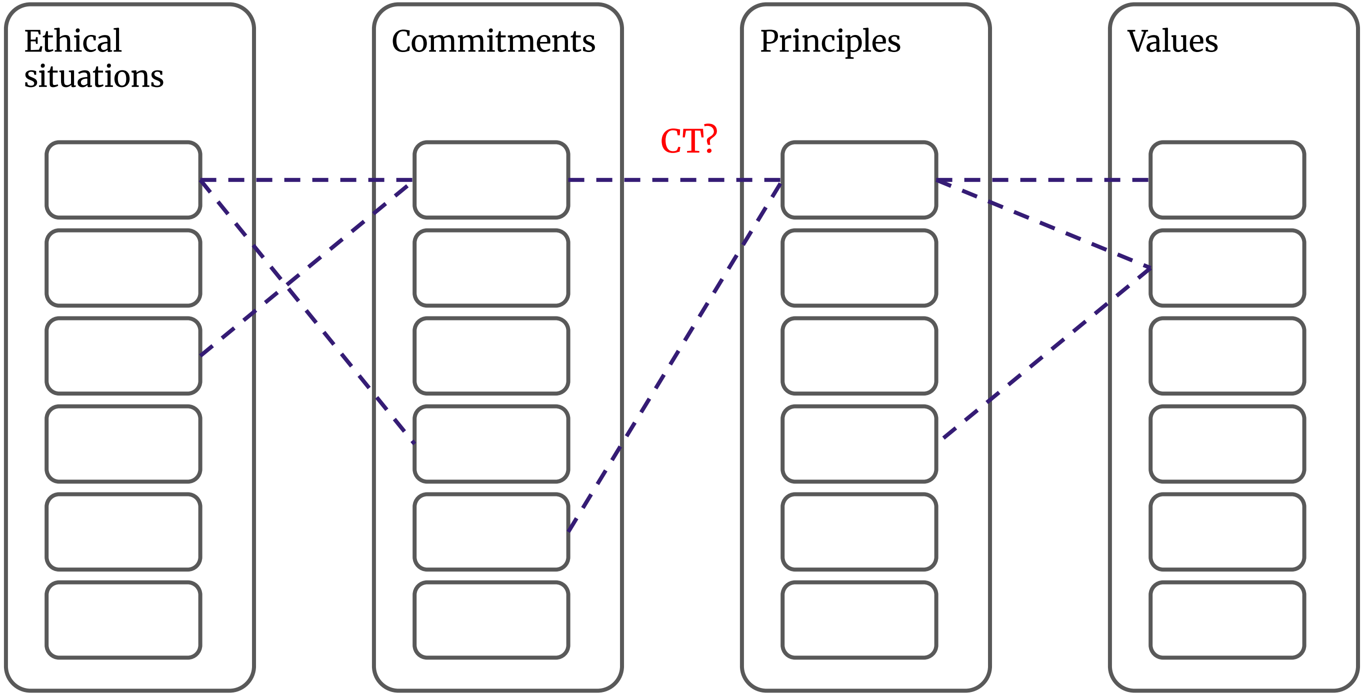
What are the axes of this space?





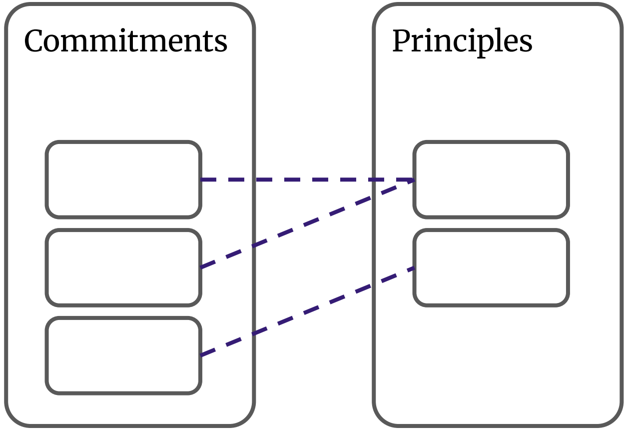
2. Draft of a framework

Framework



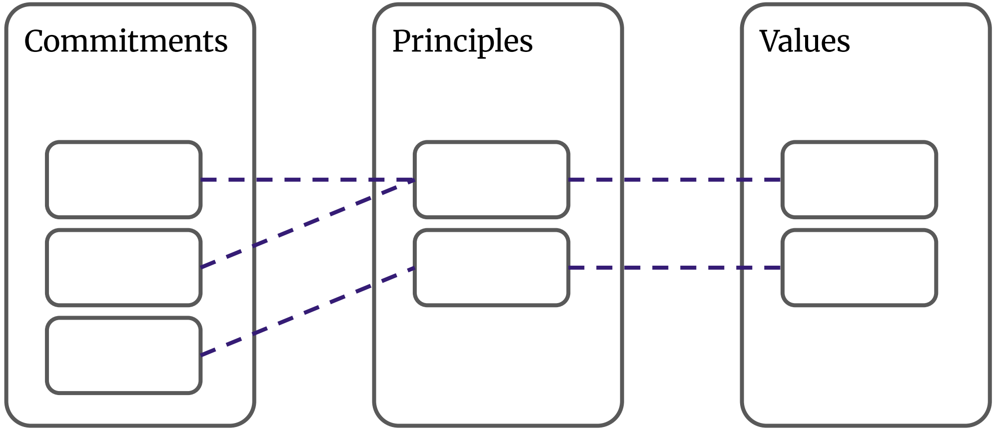
Moral reasoning

Epistemic state:

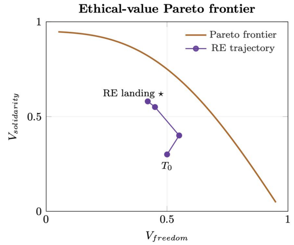


Goal: find a good epistemic state that lies on the pareto front of epistemic values and ethical values.

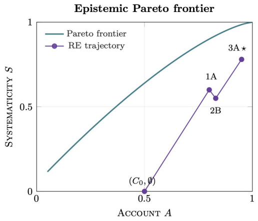
Pareto fronts



Value scoring:
Depending on which values the principles accommodate



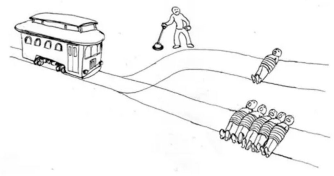
Epistemic scoring:
Account, Systematicity, Faithfulness
(Maybe others too)



3. An experiment and an idea

Naive vibe-coded RE machine

A.



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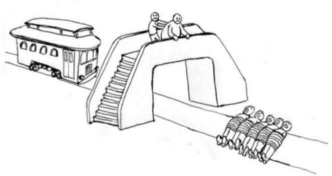
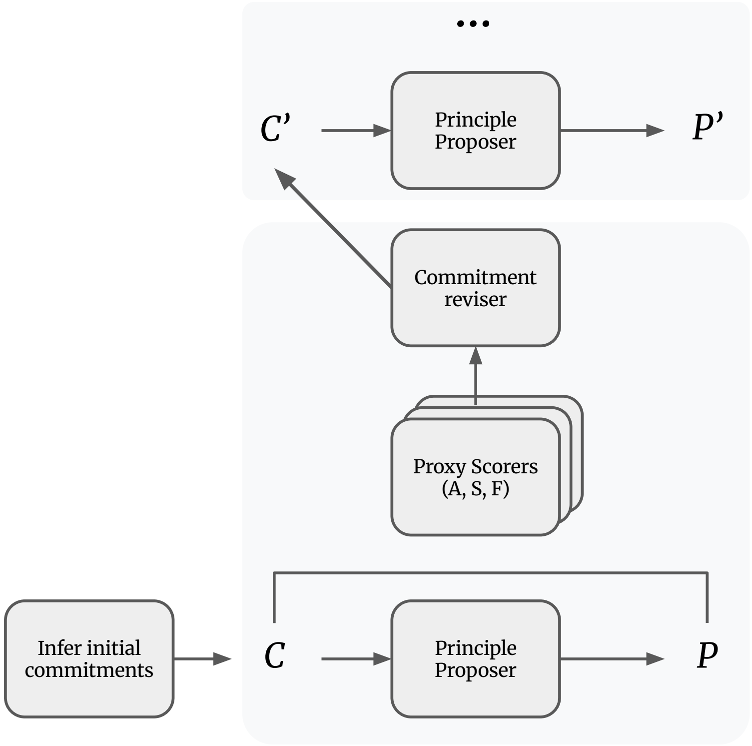




Input

Case 1 A judge in a small town presides over a case in which an angry mob has threatened to kill five hostages unless an innocent person is convicted for a crime they did not commit. The judge has the power to frame the innocent person and ensure their conviction, which would satisfy the mob and save the five hostages. If the judge refuses, the mob will kill the five hostages.

Case2 The driver of a runaway trolley with failed brakes is heading toward five workmen on the track ahead. The driver can either continue straight, in which case the five workmen will be killed, or steer onto a side track on which only one workman is working, killing that one workman instead.

Case 3 A bystander is standing next to a switch that controls the direction of a runaway trolley. If the bystander does nothing, the trolley will continue down the main track and kill five workmen. If the bystander throws the switch, the trolley will divert onto a side track on which only one workman is working, killing that one workman instead.

Case 4 A bystander is standing on a bridge over a trolley track. A runaway trolley is heading toward five workmen on the track below. The only way to stop the trolley before it reaches the five workmen is for the bystander to push a very heavy man, who is also on the bridge, down onto the track. The heavy man's body would stop the trolley and save the five workmen, but the heavy man would be killed by the impact.

Case 5 A bystander is standing next to a switch with three positions. A runaway trolley is approaching. If the bystander does nothing, the trolley will continue down the main track and kill five workmen. If the bystander throws the switch one way, the trolley will divert onto a side track and kill one workman. If the bystander throws the switch the other way, the trolley will divert onto a different side track and kill five workmen.

Commitment formation

Initial commitments (C_0):

- IC1 *(initial)* — In Case1, the judge MUST NOT frame the innocent person.
- IC2 *(initial)* — In Case2, the driver MAY divert the trolley.
- IC3 *(initial)* — In Case3, the bystander MAY throw the switch.
- IC4 *(initial)* — In Case4, the bystander MUST NOT push the heavy man.
- IC5 *(initial)* — In Case5, the bystander MAY divert the trolley to kill one.

Step 1A — Theory revision

Proposed theory T′:

- P1 — It is permissible to divert a threat from many to few.

Scoring the epistemic state

Verifier	Score	Rationale
Account	0.800	IC1=0.5: P1 is silent on framing innocent people. | IC2=1.0: P1 supports diverting the trolley to save more lives. | IC3=1.0: P1 supports diverting the trolley to save more lives. | IC4=0.5: P1 is silent on directly causing harm, even to save others. | IC5=1.0: P1 supports diverting the trolley to kill one instead of many.
Systematicity	0.600	The theory T consists of a single principle, P1, which is relatively general. However, it lacks broad implications and could potentially lead to inconsistencies if applied without considering other ethical factors, thus the score is not higher.
Faithfulness	1.000	IC1=preserved (1.0): The commitment that the judge MUST NOT frame the innocent person is held verbatim. | IC2=preserved (1.0): The commitment that the driver MAY divert the trolley is held verbatim. | IC3=preserved (1.0): The commitment that the bystander MAY throw the switch is held verbatim. | IC4=preserved (1.0): The commitment that the bystander MUST NOT push the heavy man is held verbatim. | IC5=preserved (1.0): The commitment that the bystander MAY divert the trolley to kill one is held verbatim.
Z	**0.800**	

Final state (3 steps)

Theory T:

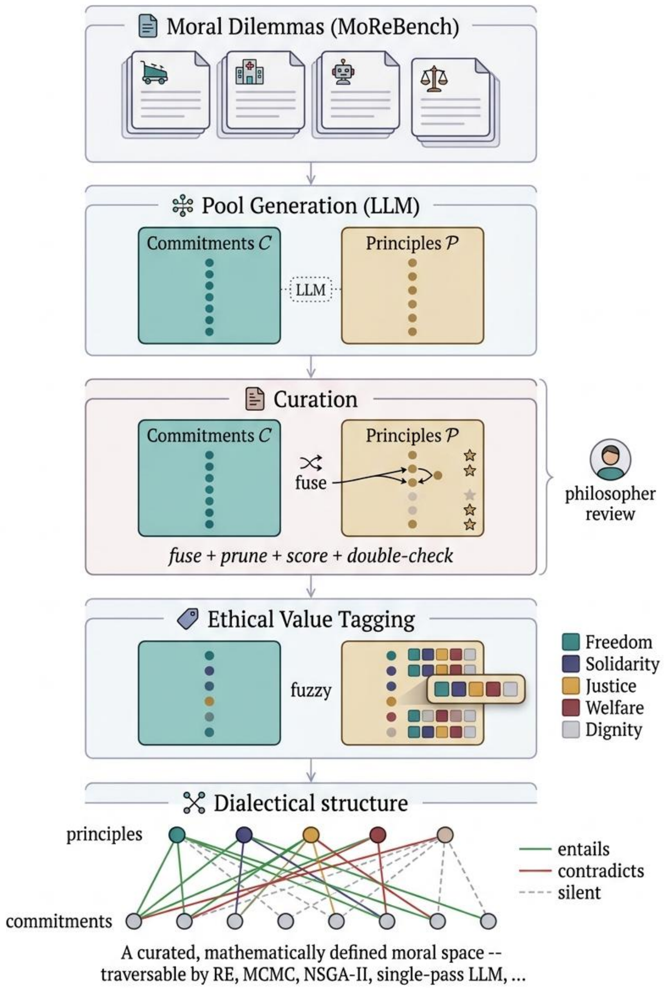
- P1 — It is permissible to divert a threat from many to few.
- P2 — It is not permissible to directly cause harm to someone to save others, unless it involves diverting an existing threat.

Commitments C:

- IC1 *(initial)* — In Case1, the judge MUST NOT frame the innocent person.
- IC2 *(initial)* — In Case2, the driver MAY divert the trolley.
- IC3 *(initial)* — In Case3, the bystander MAY throw the switch.
- IC4 *(initial)* — In Case4, the bystander MUST NOT push the heavy man.
- IC5 *(initial)* — In Case5, the bystander MAY divert the trolley to kill one.

Issues with this

- Very hard to compare different trajectories as commitments, principles and epistemic scores might warry.
- It would be nice to be able to compare different reasoning methods. E.g., RE vs other approaches (for example inspired from multi-obj opt).
- Potential solution: creating a fixed pool.



Fixed pool approach

- Given a set of dilemmas, create first a pool of commitments, principles and values.
- With inferential connections and value satisfaction.
- Generate with LLMs, select, fuse, validate by philosophers.

→ *Then we can test and compare RE as a method. And look at reasoning trajectories and pareto fronts.*

Andreas Freivogel, Does reflective equilibrium help us converge?, 2023 Synthese.

The end.